

MedSupply

Drug Shortage Early Warning & Response System

LIVE FRONTEND

<https://main.d31fsbjqraajf.amplifyapp.com>

GitHub Repository

[DHIKSHITHA0906/MedSupply](https://github.com/DHIKSHITHA0906/MedSupply)

3-Minute Demo Video

<https://youtu.be/dxzkCPR2sn0>

1. Project Overview

MedSupply is a drug-shortage early warning and response system designed to identify medicines at risk, analyze their supply-network context, evaluate sourcing interventions through simulation, and present actionable response options to a human decision-maker.

The system connects risk prediction, supply-network analysis, safe sourcing, counterfactual simulation, network-impact evaluation, optimization, and decision deadlines into a single operational workflow.

1.1 What Makes MedSupply Different

MedSupply is designed around the consequences of shortage-response decisions, not only shortage prediction. Its key differentiators are:

- (1) substitution-aware and network-aware shortage response
- (2) counterfactual simulation that evaluates whether a proposed transfer can create secondary shortages
- (3) comparison of multiple sourcing options using network risk, lead time, cost, inventory constraints and decision deadlines.

This turns the system from a simple risk dashboard into a decision-support workflow: identify the risk, understand the network, simulate intervention consequences, compare safe options, and let the human operator make the final procurement decision.

2. System Architecture

The system follows an $A \rightarrow B \rightarrow C$ pipeline followed by a serverless AWS backend and React frontend.

Layer	Responsibility
Data Sources	FDA/NDC-related and project data
Member A — Data & Risk Model	Data preparation, risk prediction, risk window, shortage handling and candidate substitution relationships
Member B — Graph & Triage	Supply-network graph, cause analysis, risk propagation and triage prioritization
Member C — Sourcing & Simulation	Safe sourcing, counterfactual simulation, network impact, optimization and decision deadlines
AWS Backend	Lambda, S3, Step Functions and API Gateway
React Frontend	Live decision-support interface deployed through AWS Amplify
Human Decision	Final procurement decision using system evidence plus real-world operational constraints

End-to-end flow: Data $\rightarrow$ A Risk Engine $\rightarrow$ B Network/Triage $\rightarrow$ C Simulation/Optimization $\rightarrow$ AWS APIs $\rightarrow$ React UI $\rightarrow$ Human Decision

3. Member A — Data & Risk Model

Member A prepares the underlying data and produces the early-warning output consumed by the downstream modules.

3.1 Key Outputs

- `risk_score` — model-derived risk indicator
- `risk_window_days` — model-derived warning
- `window_current_shortage` — FDA-derived current
- `shortage_status` candidate substitution relationships

3.2 Risk vs. Current Shortage

A data interpretation issue was identified and corrected so that predicted risk and current shortage are treated as separate signals. The risk score represents model output, while `current_shortage` represents FDA-derived current shortage status.

4. Member B — Graph & Triage Analysis

Member B analyzes the supply network surrounding a medicine and converts network information into triage and decision-support information.

- Supply-network graph
- Hospitals, warehouses and manufacturers/suppliers
- Risk propagation
- Bottleneck and cause analysis
- Triage and alert prioritization

4.1 Network Interaction

A selected medicine is displayed as the center of its supply network. Users can inspect connected suppliers, warehouses and hospitals. Selecting a facility exposes operational details and connected medicines.

5. Member C — Sourcing & Simulation

Member C evaluates potential responses to a shortage rather than presenting asingle replacement without examining network consequences.

5.1 Workflow

1. Generate candidate sources.
2. Check transferable inventory and safety constraints.
3. Simulate candidate interventions.
4. Evaluate network consequences and ripple effects.
5. Reject options that create unsafe secondary shortages.
6. Compare multiple safe response options.
7. Apply optimization and decision-deadline analysis.

5.2 Intervention Decision Support

The intervention interface allows users to compare multiple safe response options and choose one. The system provides evidence and simulation results, while the final procurement decision remains with the human operator.

5.3 Example Results — Bupivacaine

The demonstration uses Bupivacaine as a representative high-risk scenario. The model reports an 88.76% risk score, a 7-day model-derived warning window, and a current shortage status. The selected safe response uses Cencora to provide 856 units with a reported cost of ₹45,922.08 and a 2-day lead time. A cheapest-first alternative is rejected by the simulation because it creates a secondary shortage at Valley Children's.

Metric	Example
Risk score	88.76%
Warning window	7 days
Shortage status	Current
Selected source	Cencora
Selected quantity	856 units
Selected plan cost	₹45,922.08
Selected plan lead time	2 days
Rejected scenario	Cheapest-first transfer creates a secondary shortage

5.4 Pipeline-Level Output

The C pipeline used for the demonstration evaluated 33 scenarios and 165 candidate plans, with 44 candidates rejected by simulation and all 33 scenarios having at least one safe plan.

6. Frontend Workflow

LOGIN → SHORTAGES → OVERVIEW → NETWORK → INTERVENTION

6.1 Login

The user begins through a demo/operator login before entering the decision-support console.

6.2 Shortages

A dedicated Shortages view presents medicines in a compact triage list/card arrangement. Selecting a medicine reveals more information about that medicine.

6.3 Overview

The overview provides the operational picture, including:

- Active shortages
- Medicines needing action
- Medicines under watch
- Safe response plans
- Decision deadline
- Predicted stockout information

6.4 Network

The Network screen visualizes the selected medicine and its connected supply network. Clicking a supplier or facility opens a detail drawer.

- Stock
- Safety stock
- Lead time
- Status
- Connected medicines

6.5 Intervention

The Intervention screen presents multiple candidate response options. Users can compare safe options rather than being shown only one accepted plan.

- Supplier/source
- Available quantity
- Transfer quantity
- Cost
- Lead time
- Network risk
- Expiry information
- Ripple effects
- Safety/simulation status

6.6 Human Decision Boundary

A separate Human Review tab is not required in the current design. Human decision-making is incorporated into the intervention workflow. The system can identify safe options and explain why they are considered safe, but the human operator makes the final procurement decision because real-world constraints may include existing contracts, supplier preferences, budget, urgency and operational considerations.

7. AWS Deployment

The final deployment uses a serverless AWS architecture connecting the live React frontend to the backend.

AWS Service	Role
AWS Amplify	Hosts and deploys the React frontend
API Gateway	Exposes backend APIs to the frontend
AWS Lambda	Runs prediction, scenario and simulation logic
Amazon S3	Stores required data and published outputs
AWS Step Functions	Coordinates the prediction and simulation workflow

The backend endpoints were tested for prediction, scenarios and simulation. The React frontend was connected to the live AWS API, data-mapping issues were corrected, and the frontend was tested against the live backend.

7.1 Architecture Scope Decision

The final hackathon deployment intentionally uses a simplified serverless architecture rather than the broader initial architecture plan. Lambda, S3, Step Functions, API Gateway and Amplify were sufficient to deliver the working prediction, scenario and simulation workflow while reducing deployment complexity and integration risk.

8. Data Provenance & Transparency

Data Category	Examples
REAL / REAL-DERIVED	Drug identity, FDA shortage information, model-related risk output and manufacturer identities
SIMULATED	Inventory, lead time, price and expiry where generated by the simulation environment
SYNTHETIC	Network topology used for network analysis

This distinction is surfaced in the product so users can understand which evidence comes from real-derived sources and which values are generated for simulation and demonstration.

9. Safety Boundaries & Limitations

- The system is a decision-support tool and does not replace human procurement decisions.
- Candidate substitution relationships should not be interpreted as clinical interchangeability.
- The model-derived risk window is not an FDA forecast.
- Some inventory, lead-time, price and expiry values are simulated.
- Network topology may be synthetic for demonstration and analysis.
- Real-world procurement constraints may not be represented by the simulation.

10. Learning & Technical Takeaways

The project required the team to learn and integrate serverless AWS services into a working end-to-end system.

- Coordinating Lambda, S3 and Step Functions into a single workflow.
- Exposing and testing backend functionality through API Gateway.
- Deploying and connecting a React frontend through AWS Amplify.
- Integrating independently developed A, B and C modules without changing their core responsibilities.
- Diagnosing and correcting a data-semantics issue where predicted risk and current shortage status had initially been conflated.
- Rerunning downstream modules after correcting upstream data and validating that the live frontend remained consistent with the backend.
- Designing a decision-support interface that distinguishes real-derived, simulated and synthetic information.

11. Demonstration Workflow

- Login to the demo console.
- Open the Shortages view.
- Select a high-priority medicine such as Bupivacaine.
- Use Overview to understand the shortage, risk and decision deadline.
- Open Network to inspect connected suppliers and facilities.
- Select a facility to inspect stock, safety stock, lead time, status and connected medicines.
- Open Intervention to compare multiple response options.
- Inspect simulation results, network impact and ripple effects.
- Select a procurement option based on the available evidence and real-world constraints.

12. Team Contributions

Member/Role	Contributions
Katir Velavan	Data & Risk Model: FDA/NDC data, data preparation, risk prediction, risk window, shortage handling, substitution/cascade.
R. Shreya	Graph, Cause & Triage: supply-network graph, hospitals/warehouses/manufacturers, risk propagation, bottleneck analysis, common-cause & cascade analysis, harm-weighted triage.
Akshaya M V	Sourcing & Simulation: safe-source discovery, candidate generation, counterfactual simulation, network impact evaluation, weighted candidate scoring.
Dhikshitha R G	System Integration & AWS Deployment: A-> B-> C integration, Lambda, S3, Step Functions, API Gateway, API testing, live frontend/backend integration, Amplify deployment.